

1. The first and second terms of an arithmetic series are 200 and 197.5 respectively.

The sum to n terms of the series is S_n .

Find the largest positive value of S_n .

(Total 5 marks)

2. The sum of the first p terms of an arithmetic series is q and the sum of the first q terms of the same arithmetic series is p , where p and q are positive integers and $p \neq q$.

Giving simplified answers in terms of p and q , find

(a) the common difference of the terms in this series, (5)

(b) the first term of the series, (3)

(c) the sum of the first $(p + q)$ terms of the series. (3)

(Total 11 marks)

3. The angle θ , $0 < \theta < \frac{\pi}{2}$, satisfies

$$\tan \theta \tan 2\theta = \sum_{r=0}^{\infty} 2 \cos^r 2\theta$$

- (a) Show that $\tan \theta = 3^p$, where p is a rational number to be found.

(8)

- (b) Hence show that $\frac{\pi}{6} < \theta < \frac{\pi}{4}$

(2)

(Total 10 marks)

4. A sequence $\{u_n\}$ is given by

$$\begin{aligned}u_1 &= k \\u_{2n} &= u_{2n-1} \times p & n \geq 1 \\u_{2n+1} &= u_{2n} \times q & n \geq 1\end{aligned}$$

where k, p and q are positive constants with $pq \neq 1$

(a) Write down the first 6 terms of this sequence.

(3)

(b) Show that $\sum_{r=1}^{2n} u_r = \frac{k(1+p)(1-(pq)^n)}{1-pq}$

(6)

In part (c) $[x]$ means the integer part of x , so for example $[2.73] = 2$, $[4] = 4$ and $[0] = 0$

(c) Find $\sum_{r=1}^{\infty} 6 \times \left(\frac{4}{3}\right)^{\left[\frac{r}{2}\right]} \times \left(\frac{3}{5}\right)^{\left[\frac{r-1}{2}\right]}$

(4)

(Total 13 marks)

5. Given that

$$(1+x)^n = 1 + \sum_{r=1}^{\infty} \frac{n(n-1)\dots(n-r+1)}{1 \times 2 \times \dots \times r} x^r \quad (|x| < 1, x \in \mathbb{R}, n \in \mathbb{R})$$

(a) show that

$$(1-x)^{-\frac{1}{2}} = \sum_{r=0}^{\infty} \binom{2r}{r} \left(\frac{x}{4}\right)^r \quad (5)$$

(b) show that $(9-4x^2)^{-\frac{1}{2}}$ can be written in the form $\sum_{r=0}^{\infty} \binom{2r}{r} \frac{x^{2r}}{3^q}$ and give q in terms of r. (3)

(c) Find $\sum_{r=1}^{\infty} \binom{2r}{r} \times \frac{2r}{9} \times \left(\frac{x}{3}\right)^{2r-1}$ (3)

(d) Hence find the exact value of

$$\sum_{r=1}^{\infty} \binom{2r}{r} \times \frac{2r\sqrt{5}}{9} \times \frac{1}{5^r}$$

giving your answer as a rational number.

(2)

(Total 13 marks)

6. A sequence of positive integers $a_1, a_2, a_3, \dots$ has r th term given by

$$a_r = 2^r - 1$$

(a) Write down the first 6 terms of this sequence. (1)

(b) Verify that $a_{r+1} = 2a_r + 1$ (1)

(c) Find $\sum_{r=1}^n a_r$ (3)

(d) Show that $\frac{1}{a_{r+1}} < \frac{1}{2} \times \frac{1}{a_r}$ (1)

(e) Hence show that $1 + \frac{1}{3} + \frac{1}{7} + \frac{1}{15} + \frac{1}{31} + \dots < 1 + \frac{1}{3} + \left(\frac{1}{7} + \frac{1}{7} + \frac{1}{7} + \dots \right)$ (2)

(f) Show that $\frac{31}{21} < \sum_{r=1}^{\infty} \frac{1}{a_r} < \frac{34}{21}$ (5)

(Total 13 marks)

7. (a) Show that

$$\sum_{r=0}^n x^{-r} = \frac{x}{x-1} - \frac{x^{-n}}{x-1} \quad \text{where } x \neq 0 \text{ and } x \neq 1 \quad (2)$$

(b) Hence find an expression in terms of x and n for $\sum_{r=0}^n r x^{-(r+1)}$ for $x \neq 0$ and $x \neq 1$

Simplify your answer.

(4)

(c) Find $\sum_{r=0}^n \left(\frac{3+5r}{2^r} \right)$

Give your answer in the form $a - \frac{b+cn}{2^n}$, where a , b and c are integers.

(7)

(Total 13 marks)

8. [In this question the values of a , x , and n are such that a and x are positive real numbers, with $a > 1$, $x \neq a$, $x \neq 1$ and n is an integer with $n > 1$]

Sam was confused about the rules of logarithms and thought that

$$\log_a x^n = (\log_a x)^n \quad (1)$$

- (a) Given that x satisfies statement (1) find x in terms of a and n .

(3)

Sam also thought that

$$\log_a x + \log_a x^2 + \dots + \log_a x^n = \log_a x + (\log_a x)^2 + \dots + (\log_a x)^n \quad (2)$$

- (b) For $n = 3$, x_1 and x_2 ($x_1 > x_2$) are the two values of x that satisfy statement (2).

- (i) Find, in terms of a , an expression for x_1 and an expression for x_2 .

- (ii) Find the exact value of $\log_a \left(\frac{x_1}{x_2} \right)$.

(5)

- (c) Show that if $\log_a x$ satisfies statement (2) then

$$2(\log_a x)^n - n(n+1)\log_a x + (n^2 + n - 2) = 0$$

(6)

(Total 14 marks)

9. (a) The sides of the triangle ABC have lengths $BC = a$, $AC = b$ and $AB = c$, where $a < b < c$. The sizes of the angles A, B and C form an arithmetic sequence.

(i) Show that the area of triangle ABC is $ac \frac{\sqrt{3}}{4}$. (4)

Given that $a = 2$ and $\sin A = \frac{\sqrt{15}}{5}$, find

(ii) the value of b , (2)

(iii) the value of c . (4)

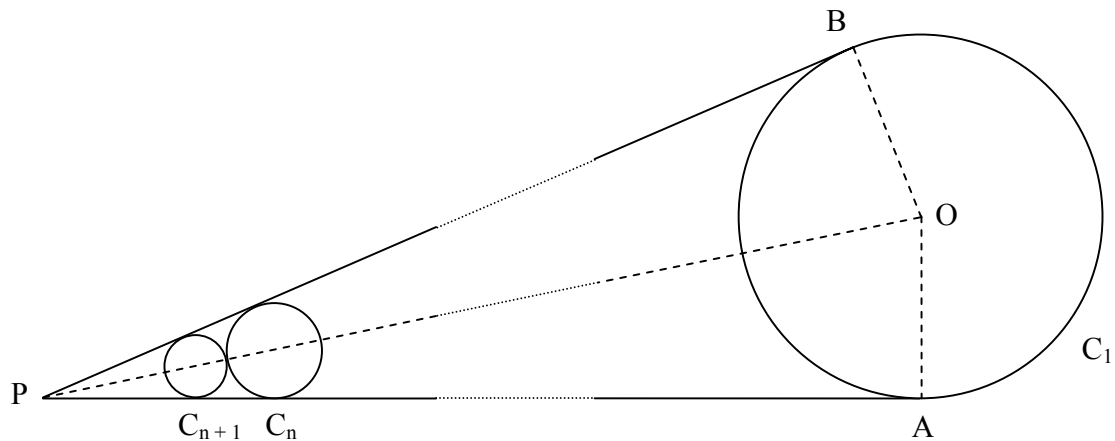
- (b) The internal angles of an n -sided polygon form an arithmetic sequence with first term 143° and common difference 2° .

Given that all of the internal angles are less than 180° , find the value of n . (5)

(Total 15 marks)

11.

Figure 2



The circle C_1 has centre O and radius R . The tangents AP and BP to C_1 meet at the point P and angle $APB = 2\alpha$, $0 < \alpha < \frac{\pi}{2}$. A sequence of circles $C_1, C_2, \dots, C_n, \dots$ is drawn so that each new circle C_{n+1} touches each of C_n , AP and BP for $n = 1, 2, 3, \dots$ as shown in Figure 2. The centre of each circle lies on the line OP .

- (a) Show that the radii of the circles form a geometric sequence with common ratio

$$\frac{1 - \sin \alpha}{1 + \sin \alpha}.$$

(5)

- (b) Find, in terms of R and α , the total area enclosed by all the circles, simplifying your answer.
- (3)

The area inside the quadrilateral $PAOB$, not enclosed by part of C_1 or any of the other circles, is S .

- (c) Show that

$$S = R^2 \left(\alpha + \cot \alpha - \frac{\pi}{4} \operatorname{cosec} \alpha - \frac{\pi}{4} \sin \alpha \right).$$

(5)

- (d) Show that, as α varies,

$$\frac{dS}{d\alpha} = R^2 \cot^2 \alpha \left(\frac{\pi}{4} \cos \alpha - 1 \right).$$

(4)

- (e) Find, in terms of R , the least value of S for $\frac{\pi}{6} \leq \alpha \leq \frac{\pi}{4}$.
- (3)

12.

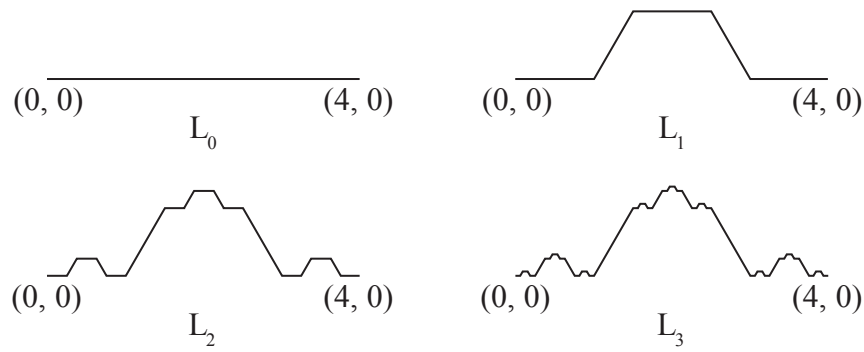


Figure 2

Figure 2 shows the first few iterations in the construction of a curve, L .

Starting with a straight line L_0 of length 4, the middle half of this line is replaced by three sides of a trapezium above L_0 as shown, such that the length of each of these sides is $\frac{1}{4}$ of the length of L_0

After the first iteration each line segment has length one.

In subsequent iterations, each line segment parallel to L_0 similarly has its middle half replaced by three sides of a trapezium above that line segment, with each side $\frac{1}{4}$ the length of that line segment.

Line segments in L_n are either parallel to L_0 or are sloped.

(a) Show that the length of L_2 is $\frac{23}{4}$ (2)

(b) Write down the number of

- (i) line segments in L_n that are parallel to L_0
- (ii) sloped line segments in L_2 that are not in L_1
- (iii) **new** sloped line segments that are created by the $(n + 1)$ th iteration. (3)

(c) Hence find the length of L_n as $n \rightarrow \infty$ (6)

The area enclosed between L_0 and L_n is A_n

(d) Find the value of A_1 (2)

(e) Find, in terms of n , an expression for $A_{n+1} - A_n$ (3)

(f) Hence find the value of A_n as $n \rightarrow \infty$ (3)

Question 12 continued

The same construction as described above is applied externally to the three sides of an equilateral triangle of side length a .

Given that the limit of the area of the resulting shape is $26\sqrt{3}$

(g) find the value of a .

(3)

(+S2)